MIL-F-14988(MU)
30 December 1968

SUPERSEDING
PA-PD-2349
6 September 1963

MILITARY SPECIFICATION

FIRING AND SAFING SQUIB-SWITCH MODULE

ASSEMBLIES

1. SCOPE

1.1 This specification covers special requirements not included on the drawings, and provides the quality assurance provisions for the fabrication of parts, assembling, nd the preparation for delivery for one type of firing and safing squib switch assemblies used on Atomic Demolition charge and designated as firing and safing squib-switch module assemblies.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-490 - Cleaning and Preparation of Ferrous and Zinc Coated Surfaces for Organic Protective Coating.

MIL-S-14987(MU) Squib Switch

MIL-S-60902(MU) Soldering Connections for Nuclear Weapons Systems, Procedure for.

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-1168(MU) Lot Numbering of Ammunition

MIL-STD-1169(MU) Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC 1125

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DRAWINGS

ORDNANCE CORPS
8823336 - Firing Squib-Switch Module Assembly.
8823337 - Safing Squib-Switch Module Assembly.
8867511 - Functional Test Panel.
8867512 - Functional Test, Operating Instructions for.
EL-8823336 - Equipment List
EL-8823337 - Equipment List

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material used to fabricate any part complies with the specifications listed on the applicable drawing.

3.2 Assembly.

3.2.1 Squib switches. Records shall be furnished showing that the squib switches have been tested, inspected and accepted in accordance with MIL-S-14807(MU).

3.2.2 Firing squib-switch module assembly. The assembly shall comply with all requirements specified on Drawing (dwg) 8823336 and with all requirements specified in applicable specifications.

3.2.3 Safing squib-switch module assembly. The assembly shall comply with all requirements specified on dwg. 8823337 and with all requirements specified in applicable specifications.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

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3.4 Continuity and resistance. The assembly shall have continuity and resistance as specified in the applicable table when tested as specified in 4.4.1.

Table I
Continuity and resistance
Firing Squib-switch module assembly (see dwg 8823336)
before functioning

Pin	to	Pin	Value
3		20	2.45 ± 0.27 ohms
5		20	2.45 ± 0.27 ohms
6		20	2.45 ± 0.27 ohms
7		20	2.45 ± 0.27 ohms
14*		20	2.45 ± 0.27 ohms
17*		20	2.45 ± 0.27 ohms
19		20	2.45 ± 0.27 ohms
21		20	2.45 ± 0.27 ohms
8		20	0.050 maximum
18		10	0.135 maximum

Table II
Continuity and resistance
Firing squib-switch module assembly (see dwg 8823336)
after functioning

Pin	to	Pin	Value
3		7	0.050 maximum
3		9	0.050 maximum
3*		17	0.050 maximum
3		21	0.050 maximum
6		5	0.050 maximum
6*		14	0.050 maximum
6		19	0.050 maximum
6		22	0.050 maximum
8		20	0.050 maximum

* Fuze clip shorted

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Table III
Continuity and resistance
Firing squib-switch module assembly (see dwg 8823336)
after removal of potting

Note: Prior to this test, disconnect jumpers between S106-2A and S105-2B; S104-2A and S103-2B; S102-2A and S101-2A and S202-2B; S203-2A and S204-2B; and S205-"a and S206-2B.

Switch terminal to	Switch terminal	Value (ohms)
S106-2A	S106-2B	0.030 max.
S105-2A	S105-2B	0.030 max.
S104-2A	S104-2B	0.030 max.
S103-2A	S103-2B	0.030 max.
S102-2A	S102-2B	0.030 max.
S101-2A	S101-2B	0.030 max.
S206-2A	S206-2B	0.030 max.
S205-2A	S205-2B	0.030 max.
S204-2A	S204-2B	0.030 max.
S203-2A	S203-2B	0.030 max.
S202-2A	S202-2B	0.030 max.
S201-2A	S201-2B	0.030 max.

Table IV
Continuity and resistance
Safing squib-switch module assembly (see dwg 8823337)
before functioning

Pin	to	Pin	Value, (ohms)
1		2	0.060 max.
3		16	0.060 max.
7		19	0.060 max.
8		12	0.060 max.
8		13	0.060 max.
11		10	0.060 max.
11		22	0.060 max.
11		23	0.060 max.
20		24	0.060 max.
20		25	0.060 max.
3		19	1.10 ± 0.30 max

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Table V
Continuity and resistance
Safing squib-switch module assembly (see dwg 8823337)
after functioning

Pin	to	Pin	Value, Ohms
1		15	0.060 ohms, max.
6		11	0.060 ohms, max.
6		23	0.060 ohms, max.
7		19	0.060 ohms, max.
10		22	0.060 ohms, max.
12		13	0.060 ohms, max.
16		3	0.060 ohms, max.
24		25	0.060 ohms, max.
Shell		19	100 ohms, min.
Shell		3	100 ohms, min.

Table VI
Continuity and resistance
Safing squib-switch module assembly (see dwg 8823337)

Note: Prior to this test, disconnect jumper between S301-4B and S302-4A or S301-4A and S302-4B.

Switch-terminal	to	Switch terminal	Value Ohms
S302-4A		S302-4B	0.030 max.
S301-4A		S301-4B	0.030 max.

3.5 Insulation resistance. The assembly having complied with the requirements of 3.4, shall have values of insulation resistance as specified in the applicable table at a potential of 500 ± 50 volts direct current (vdc) with current limited to 5 milliamperes (ma) when tested as specified in 4.4.2.

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Table VII
Insulation resistance
Firing Squib-Switch module assembly (see dwg 8823336)
before functioning

Pin (+)	to	Pin (-)	Exceptions	Value (megohms)
9		All other pins	None	20.0 min.**
14*		All other pins	None	20.0 min.**
17*		All other pins	None	20.0 min.**
18		All other pins	Pin 10	20.0 min.**
22		All other pins	None	20.0 min.**
6*		All other pins	Pins 3-8-21-7-20-19-5	20.0 min.**
Shell & Sleeves		All other pins	None	20.0 min.**

* Fuse removed.
** After the humidity test, the value of megohms shall be 1.0 minimum.

Table VIII
Insulation resistance
Firing squib-switch module assembly (see dwg 8823336)
after functioning

Pin (+)	to	Pin (-)	Exceptions	Value (megohms)
3*		All other pins	Pins 9-21-7	20.0 min.**
6*		All other pins	Pins 5-19-22	20.0 min.**
8		All other pins	Pin 20	20.0 min.**
10		All other pins	None	20.0 min.**
14*		All other pins	None	20.0 min.**
17*		All other pins	None	20.0 min.**
18		All other pins	None	20.0 min.**
Shell & Sleeves		All other pins	None	1.0 min.

* Fuse removed.
** After the humidity test, the value of megohms shall be 1.0 minimum.

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Table IX
Insulation resistance
Safing squib-switch module assembly (see dwg 8823337)
before functioning

Checks J42-6	to All Other Pins	Except Pins	Values
2	"	1	20 meg. ohms, min.*
6	"	-	20 meg. ohms, min.*
11	"	10-22-23	20 meg. ohms, min.*
13	"	8-12	20 meg. ohms, min.*
15	"	-	20 meg. ohms, min.*
25	"	20-24	20 meg. ohms, min.*
3-7-16-19	"	-	20 meg. ohms, min.*
Shells & sleeves	"	-	20 meg. ohms, min.*

* After the humidity test, the value of megohms shall be 1.0 minimum.

Table X
Insulation resistance
Safing squib-switch module assembly (see dwg 8823337)
after functioning

Checks J42-6	to All Other Pins	Except Pins	Values
2	"	-	20 meg. ohms, min.*
3	"	7-16-19 Shells & sleeves	20 meg. ohms, min.*
8	"	-	20 meg. ohms, min.*
11	"	6-23	20 meg. ohms, min.*
13	"	12	20 meg. ohms, min.*
15	"	1	20 meg. ohms, min.*
19	"	3-7-16 Shell & sleeves	20 meg. ohms, min.*
20	"	-	20 meg. ohms, min.*
22	"	10	20 meg. ohms, min.*
25	"	24	20 meg. ohms, min.*
Shells & sleeves	"	3-7-16-19	20 meg. ohms, min.

* After humidity test, the value of megohms shall be 1.0 minimum.

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Table XI
Insulation resistance
Safing squib-Switch module assembly (see dwg 8823337)
after Potting is Removed

Switch-Terminal	to	Switch-terminal	Value
S303-4A		S303-4B	20.0 megohms, min.
S304-4A		S304-4B	20.0 megohms, min.

3.6 Functioning. - The assembly, having complied with the requirements of 3.4 and 3.5, shall function in the time specified in the applicable table and after functioning shall comply with the requirements of 3.4 and 3.5 when tested as specified in 4.4.3 in addition, upon completion of all tests the potting shall be removed from the assembly and the assembly shall be inspected as specified in accordance with the applicable table. The individual switches shall then be sectioned and there shall be no evidence of potting seepage past the inner boss located approximately 1/8 inch below the crimp that secures the contact holder within the housing.

Table XII
Functioning
Firing Squib-Switch Module assembly (see dwg 8823336)

1.	Input Voltage:	15 ± 3 Volts, direct current			
2.	Supply	6 amp. min.			
3.	Functioning	Indicated by Current to			
	Time (seconds)	Bank 1	S101 S106	Bank 2	S201 S206
	T_o - Application of Voltage to	Pins 3 (+) and 8 (-)		Pins 6 (+) and 20 (-)	
	T_1 - 1 ± 50 percent after T_o	Pins 3 and 7		Pins 6 and 19	
	T_2 - 8 ± 25 percent after T_1	Pins 3 and 21		Pins 6 and 5	
	T_3 - 3 ± 40 percent after T_2 (room temperature)	Pins 3 and 9		Pins 6 and 22	
	3 ± 25, -40 percent after T_2 (± 165°F.)				
	3 ± 50, -0 percent after T_2 (-65°F.)				

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Table XIII
Functioning
Safing Squib-switch module assembly (see dwg 8823337)

1.	Input Voltage:	15 ± 3 volts, direct current
2.	Supply	6 Amp. min.
3.	Functioning	Indicated by Current to
	Time (.010 seconds, max.)	Pins
T_o	Application of voltage to	3 (+) and 16 (+); 7 (-) and 19 (-)
T_1	Continuity (.010 seconds max. after T_o	6 and 11; 6 and 23 or 1 and 15
T_1	Discontinuity (.010 seconds max. after T_o)	10 and 11; 11 and 22 or 1 and 2

3.7 High temperature.- The assembly having complied with the requirements of 3.4 and 3.5 shall withstand exposure to a temperature of 160 plus or minus (±) 5 degrees Fahrenheit (°F), without swelling, bubbling, cracking or other deterioration and shall comply with the requirements of 3.4, 3.5 and 3.6 when tested as specified in 4.4.4.

3.8 Low temperature. - The assembly, having complied with the requirements of 3.4 and 3.5, shall withstand exposure to a temperature of -65 ± 5°F without cracking, embrittlement or other deterioration and shall comply with the requirements of 3.4, 3.5 and 3.6 when tested as specified in 4.4.6.

3.9 Humidity.- The assembly, having complied with the requirements of 3.4 and 3.5, shall withstand exposure to a relative humidity of 90 to 95 percent, without swelling, bubbling, cracking or other deterioration and shall comply with the requirements of 3.4, 3.5 and 3.6 when testes as specified in 4.4.7.

3.10 Marking adhesion. - The marking of the assembly shall show no evidence of peeling, flaking or other deterioration when tested as specified in 4.4.8.

3.11 Reworking. - No assembly shall be reworked after being inspected in accordance with the examination specified in 4.3.2.1.1 unless the examination specified in 4.3.2.1.1 is completely reconducted.

3.12 Serialization.

3.12.1 Not required on drawings. - The contractor may assign serial numbers for internal control purposes at his descretion. The finish or any dimension of the item shall not be altered or damaged by the application of the serial number. The material used for marking shall not be subject to corrosion, fungus, flaking, or other similar adverse action which could prevent proper functioning of the item.

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3.12.1.1 Unclassified items. - The contractor may select any number suitable for his purpose.

3.12.1.2. Classified items. - Production quanities shall not be revealed. Random serial numbers shall be selected by the contractor in a manner which has been approved by the procuring agency. These numbers shall not be duplicated.

3.12.2 Required by drawing. - The contractor shall assign serial numbers as required by drawings.

3.12.2.1 Unclassified items. - The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.12.2.2 Classified items. - Production quanities shall not be revealed. Random serial numbers shall be selected by the contractor in a manner which has been approved by the procuring agency. These numbers shall not be duplicated.

3.13 Workmanship. -

3.13.1 Plastic and metal characteristics. - All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions,porosity, or any similar characteristic.

3.13.2 Threads. - Threads shall be full and undamaged for the entire minimum.length or depth as required on the applicable drawing.

3.13.3 Burrs. - Parts shall be free from burrs.

3.13.4 Foreign matter. - Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.13.5 Cleaning. - The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.13.6 Electrical. - Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4 QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections

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set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 each complete assemblies (see dwgs 8823336 and 8823337) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in table XIV Tests shall be conducted in order specified in table XIV.

Table XIV

Order of test	Number of samples subjected to test	Number of samples subjected to functioning during or after referenced test
1. Classification of Characteristics (see 4.3.2.1)	3	
2. Continuity & Resistance (see 3.4)	3	-
3. Insulation resistance (see 3.5)	3	-
4. Functioning (3.6)	3 (see next column)	-
5. High temperature (see 3.7)	3	1
6. Low temperature (see 3.8)	2	1
7. Humidity (see 3.9)	1	1
8. Marking adhesion (see 3.10)	3	-

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number.

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procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot.- An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. - Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Major characteristic table II

Lot size	Sample size
1- 155	100% inspection
156- 200	155
201- 250	175
251- 300	190
301- 350	200
351- 400	210
401- 450	220
451- 500	225
501- 550	230
551- 600	235
601- 650	240
651- 700	243
701- 750	247
751- 800	250
801- 900	254
901-1000	258
1001-1100	261
1101-1200	264
1201-1300	266
1301-1400	268
1401-1500	270
1501-1600	272

Minor characteristic table III

Lot size	Sample size
1- 25	100% inspection
26- 30	25
31- 35	26
36- 40	27
41- 45	28
46- 50	29
51- 55	30
56- 60	31
61- 65	32
66- 70	33
71- 75	34
76- 85	35
86- 95	36
96-110	37
111-130	38
131-156	39
157-190	40
191-240	41
241-340	42
341-520	43
521-1120	44
1121 and over	45

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4.3.2.1 Classification of characteristics. - Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Module assembly, prior to potting (see dwgs 8823336 and 8823337)

Categories	Characteristics	Method of inspection
Critical:		
1.	Proper wiring, including routing	Visual and Wire List
2.	Soldering in accordance with MIL-S-60902(MU)	Visual/Manual
3.	No component damaged	Visual
4.	Correct squib-switch	Visual
5.	Sealing compound present, sufficient, or properly applied to crimp of housing around plastic contact holder	Visual
6.	Glass tape present or properly applied (as applicable)	Visual
Major:		
101.	Sleeving present	Visual
102.	No component incorrectly assembled or positioned	Visual/Manual
103.	Workmanship (see 3.13)	Visual/Manual

Minor: None defined.

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4.3.2.1.2 Module assembly after potting (see dwgs. 8823336 and 8823337)

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Location of connector	Gage
102.	Location of sleeve	Gage
103.	Height	Gage
104.	Width	Gage
105.	Length	Gage
106.	Height of fuse mount (if applicable)	Gage
107.	Depth of connector	Gage
108.	No Sleeve missing or above flush (2)	Visual/Manual
109.	No Cracks, voids or bubbles in potting	Visual/Manual
110.	Marking	Visual
111.	Workmanship (see 3.13)	Visual/Manual
112.	No connector pin damaged	Visual
*113.	Length of connector pins do not differ by more than 1/16 inch	Visual/Gage
114.	Connector insert not damaged	Visual

* In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, a gage shall be used to determine the variation from the normal height. Reject any pin that varies by more than .063 inch max. from the normal height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from the measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements of pin height.

4.3.2.1.3 Preparation for delivery (see section 5).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Packaging, packing and marking in accordance with MIL-STD-1169(MU)	Visual
Minor:		
201.	Workmanship (see 3.13)	Visual/Manual

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4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

Test	Test classification
a. Continuity & resistance	major
b. Insulation resistance	major
c. High temperature	major
d. Low temperature	major
e. Marking adhesion	minor

Of these tests, c and d are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Continuity and resistance (table I and Table IV) (see 3.4), insulation resistance (table VII and table IX) (see 3.5) and marking adhesion (see 3.10). Each assembly in the lot shall be subjected to these tests in the order specified (see applicable paragraph 4.4). If any assembly fails to comply with the applicable requirements, it shall be classified defective and removed from the lot.

4.3.3.2 High temperature (see 3.7), low temperature (see 3.8). Sample assemblies shall be selected from each lot in accordance with table XV and subjected to test, (see applicable paragraph in 4.4). For lots of 300 or less, if any assembly fails to comply with the applicable requirements, the lot shall be rejected.

Table XV

Lot Size	Sample Size
2-8	1
9-15	2
16-25	3
26-40	5
41-65	8
66-110	13
111-180	22
181-300	36
301 and over	MIL-STD-105 (AQL 0.25 percent)

4.3.4 Acceptance inspection equipment.

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4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL-8823336 and EL-8823337. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity and resistance (See 3.4). Continuity and resistance shall be measured with approved resistance measuring equipment.

4.4.2 Insulation resistance (See 3.5). Insulation resistance shall be measured with approved insulation resistance equipment.

4.4.3 Functioning (See 3.6). The functioning test shall be conducted utilizing test panel dwg 8867511 and operating instructions 8867512. After functioning, the assembly shall be tested as specified in 4.4.1 and 4.4.2. Testing to determine functioning of individual switches shall be conducted with approved resistance measuring equipment to measure resistance values and approved insulation resistance equipment to measure insulation resistance values.

4.4.4 High temperature (See 3.7). The assembly shall be placed in a chamber and the temperature within the chamber shall be raised to +160 $\pm$ 5°F. After stabilization (see 6.4) the assembly shall be exposed to the temperature of +160°F. for a period of not less than 4 hours. At the end of the time period, the assembly shall be tested as specified in 4.4.1, 4.4.2 and 4.4.3 at +160 $\pm$ 5°F without removing from chamber.

4.4.5 Low temperature. The assembly shall be placed in a chamber and the temperature within the chamber shall be reduced to -65 $\pm$ 5°F. After stabilization (see 6.4), the assembly shall be exposed to the temperature of -65°F. for a period of not less than 4 hours. At the end of the time period, the assembly shall be tested as specified in 4.4.1, 4.4.2 and 4.4.3 at -65 $\pm$ 5°F without removing from the chamber.

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4.4.6 Humidity. - The assembly shall be placed in a chamber and the temperature within the chamber shall be raised to +160 ± 5°F. After stabilization (see 6.4), the assembly shall be exposed to the temperature of +160°F. for a period of 2 hours and a relative humidity of 90 to 95 percent shall be maintained while the chamber temperature is +160°F. At the end of the 2 hour period, the chamber temperature shall be uniformly decreased, during a 1 hour period to between + 68°F. and + 95°F. During the period of decreasing temperature, the relative humidity shall be 100 percent. The assembly shall be subjected to ten such humidity cycles (1 hour of increasing temperature at 90 to 95 percent relative humidity, 2 hours of temperature stabilization at 90 to 95 percent relative humidity, and 1 hour decreasing temperature at 100 percent relative humidity). The water used in controlling the relative humidity shall be distilled, having a pH value of between 6.5 and 7.5 at + 77°F. Upon completion of the tenth cycle, the assembly shall be immediately removed from the test chamber and dried by blowing dry, filtered, compressed air over the assembly for not more than 5 minutes. The assembly shall then be tested as specified in 4.4.1, 4.4.2 and 4.4.3.

4.4.7 Marking adhesion. - The adhesion of each marked surface shall be determined by the tape test specified in Specification MIL-C-490.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

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6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Temperature stabilization. The temperature is considered stabalized when the greatest peaks (both highest and lowest) continually observed over 5 minutes do not exceed the nominal temperature plus the tolerance specified.

Custodian:
Army MU

Preparing Activity
Army-MU
Project No. 1125-A002

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-F-14988 Firing and Safing Squib-Switch Module

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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